STAT 230A Final Project Proposal

Predicting Surface Fuel Loads from Terrestrial LiDAR Scans

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Introduction

Surface fuels (dead woody material on the forest floor) are a critical forest feature to measure and monitor over time because they are a primary driver of wildfire behavior. Surface fuel measurements are essential for assessing wildfire risk and for evaluating the success of fuel reduction treatments. Surface fuels are highly variable across time and space and, therefore, must be measured frequently at high intensity. Unfortunately, they are extremely labor intensive to measure using the current standard field-based methods.

A promising alternative method is to predict surface fuel loads from terrestrial LiDAR scans. Terrestrial LiDAR is a ground-based remote sensing technology that generates detailed 3D point clouds (Fig. S1). A single terrestrial LiDAR scan only takes about 5 minutes in the field, which is substantially faster than the current standard methods. Our goal is to build a model to predict surface fuel loads from metrics derived from terrestrial LiDAR scans.

Data description

184 plots were measured across six sites in the Sierra Nevada mixed-conifer forest zone (Table 1). The plots from site BLANK were measured and provided by Kea Rutherford. The data from the rest of the sites were provided by Kea's research collaborators at UC San Diego.

site	number of plots
BGL	26
DLB	27
IND	32
SHA	25
TCU	49
WIN	25
all sites	184

Table 1: Table 1. Number of observations across sites.

At each plot, a terrestrial LiDAR scan was taken at plot center. All scans were uploaded to and processed by IntELiMon (Interagency Ecosystem LiDAR Monitoring program maintained by the US Geological Survey's Earth Resources Observation and Science Center). The output from IntELiMon is a large set of plot-level summary metrics derived from the terrestrial LiDAR scans. Some example metrics include: (1) fine_l1_cnt (count of points in the 0-3 meter high voxelized point clouds with tree stems and shrubs removed), (2) TBA (total basal area of all detected overstory trees calculated from classified stem points in the point cloud), and (3) shrubs_l1_cnt (count of points in the Shrub classified voxelized point cloud).

At each plot, surface fuels were also directly measured using line-intercept transects (which is the current standard field-based method; Brown 1974). Surface fuel loads were then calculated from the line-intercept transect data using appropriate equations developed for Sierra Nevada mixed-conifer forests (van Wagtendonk et al. 1996, 1998).

For this project, the response variable is total fine surface fuel load in megagrams per hectare (fine_load_Mg_ha) calculated from the line-intercept transect data. Fine surface fuels are defined as ≤ 3 inches in diameter. This response variable is continuous, ranging from 0.89 to 39.48. The predictor variables are 75 summary metrics derived by IntELiMON from terrestrial LiDAR scans. All the predictor variables are either continuous or positive integers.

```
data <- read_csv("fuels_data.csv", show_col_types = FALSE)
head(data)
```

```
## # A tibble: 6 x 77
##   site  plot  fine_load_Mg_ha vox_l1_cnt vox_l1_mean vox_l1_median vox_l1_std
##   <chr> <chr>           <dbl>      <dbl>       <dbl>         <dbl>      <dbl>
## 1 BLG   101             20.3     2108947        8.22          6.17       6.73
## 2 BLG   102             10.6     1469968        4.96          3.14       5.53
## 3 BLG   104              4.82    1413916        3.19          2.90       2.61
## 4 BLG   105              6.95    1620433        6.07          3.93       6.37
## 5 BLG   106             15.6     2296114        9.11          9.18       6.53
## 6 BLG   107              4.43    1732248        3.26          2.59       3.15
## # i 70 more variables: vox_l1_tgi <dbl>, vox_l1_vari <dbl>, vox_l1_skew <dbl>,
## #   vox_l1_kurt <dbl>, TBA <dbl>, Basalarea <dbl>, MeanTH <dbl>, MDBH <dbl>,
## #   TreesN <dbl>, MaxTH <dbl>, SDHT <dbl>, CBH <dbl>, LAI <dbl>, ULAI <dbl>,
## #   MLAI <dbl>, OLAI <dbl>, FHD <dbl>, GiSimp <dbl>, LAHV <dbl>,
## #   fuel0_3l1_cnt <dbl>, fuel0_3l1_mean <dbl>, fuel0_3l1_median <dbl>,
## #   fuel0_3l1_std <dbl>, fuel0_3l1_tgi <dbl>, fuel0_3l1_vari <dbl>,
## #   fuel0_3l1_skew <dbl>, fuel0_3l1_kurt <dbl>, shrubs_l1_cnt <dbl>, ...
```

Regression analysis Plan

Data preprocessing

Prior to modeling, all predictor variables will be scaled and centered to ensure that the LASSO penalty is appliet equitably across variables, regardless of their original units.

Train/Test Split

The dataset sill be split into training (75%) and test (25%) sets. Because obserations were collected across 6 distinct field sites, we will perform a stratified random split, sampling proportionally from each site. This ensures that all sites are represented in both the training and test sets and prevents site-level imbalance from inflating or deflating model performance estimates.

** add in exact number of scanes per site? **

Model: LASSO regression

We will fit a LASSO (Least Absolute Shrinkage and Selection Operator) regression model to predict [response variable] from the [80] voxelized LiDAR features. LASSO was chosen over ridge regression because it performs variable selection by shrinking coefficients of irrelevant predictors to exactly zero. Given the large number of predictors relative to observations and the expectation that a subset of voxel features will be sufficient to explain variation in the response, LASSO's sparsity-including penalty is preferable. Additionally, several features are represented by both their mean and median, introducing multicollinearity among predictors. Unlike ridge regression, which retains all correlated variables with shrunk coefficients, LASSO's tendency to select one variable from a correlated group and zero out the others is preferable here, as we do not require both the mean and median of a feature to be retained in the final model.

Hyperparameter Tuning

The regularization parameter λ controls the degree of shrinkage and will be selected via k-fold cross-validation on the training set. We will evaluate the model perfomance across a grid of λ values and select the value that minimizes cross-validated mean squared error (CV-MSE). The selected λ will then be used to refit the model on the full training set.

** add exact range of lambda values? **

Model Diagnositics

After fitting the final model we will assess the following:

- Linearity and homoskedasticity: Using a residuals vs. fitted values plot, we can check for non-linear patterns and unequal variance. If heteroskedasticity is detected, we will consider a variance-stabilizing transformation of he response or the use of heteroskedasticity-robust standard errors.
- Normality of residuals: A Q-Q plot of residuals will be inspected to assess whether the normality assumption is approximately satisfied.

Evaluation

Final model performance will be evaluated on the held-out test set using room mean squared error (RMSE) and R^2